

BRIGNOGAN, France

WMO: 071070

Lat: 48.676N

Lon: 4.331W

Elev: 59

StdP: 14.66

Time Zone: 1.00 (EUC)

Period: 99-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	34.3	36.8	26.4	20.4	37.5	29.1	23.1	39.2	46.4	45.7	42.9	45.9	11.0	110	0.688

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.0	71.1	63.6	68.8	62.8	67.2	62.2	65.3	68.9	64.1	67.3	63.1	66.1	13.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	88.8	66.8	62.7	85.4	65.8	61.7	82.4	64.9	30.2	69.1	29.3	67.5	28.6	66.3	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
38.4	33.2	29.3	30.6	80.2	2.2	4.4	29.0	83.4	27.7	86.0	26.4	88.5	24.8	91.7
			28.6	68.2	2.3	1.5	26.9	69.2	25.6	70.1	24.3	70.9	22.6	72.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.2	47.0	46.6	48.5	51.1	55.3	59.2	61.9	62.3	60.6	56.9	51.9	48.1
	DBStd	6.77	4.54	4.28	4.07	3.29	2.95	2.82	2.49	2.40	3.13	3.90	3.78	4.78
	HDD50	434	110	108	76	24	1	0	0	0	0	2	22	91
	HDD65	3977	557	516	512	418	302	176	100	91	138	252	392	523
	CDD50	1955	18	12	29	56	164	277	370	381	318	217	80	33
	CDD65	21	0	0	0	0	0	2	5	7	6	1	0	0
	CDH74	44	0	0	0	0	2	9	4	14	12	3	0	0
	CDH80	7	0	0	0	0	0	2	0	2	2	1	0	0
(14) Wind	WSAvg	14.1	16.7	17.0	16.0	14.6	13.7	11.9	11.2	10.8	11.8	12.7	16.2	16.6
(15) Precipitation	PrecAvg	39.9	4.8	4.0	2.5	3.0	2.4	2.1	2.4	2.3	2.0	4.3	5.0	4.8
	PrecMax	48.1	8.8	8.3	6.9	6.8	4.9	5.2	5.1	5.7	5.1	6.7	9.6	9.2
	PrecMin	31.5	1.1	0.6	0.4	0.6	0.5	0.3	0.4	0.5	0.3	0.9	1.1	0.8
	PrecStd	5.0	2.1	1.9	1.6	1.8	1.3	1.3	1.3	1.5	1.2	1.4	2.1	2.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.0	57.6	62.8	64.7	69.0	71.2	73.9	75.5	74.0	70.6	62.8	59.6
		MCWB	54.2	53.6	54.0	55.1	59.7	63.1	64.0	66.0	64.9	63.5	59.0	55.4
	2%	DB	55.7	55.3	57.6	59.7	63.8	67.3	69.8	70.1	69.8	66.2	60.5	57.6
		MCWB	53.7	52.5	52.9	53.7	57.7	61.4	63.3	63.4	63.3	61.2	57.2	55.0
	5%	DB	54.6	53.9	55.6	57.7	61.3	65.4	68.0	67.9	67.1	64.0	58.6	56.1
		MCWB	52.6	51.5	52.4	53.0	56.8	60.4	62.8	62.9	62.5	60.2	55.8	53.9
	10%	DB	53.3	52.7	53.9	56.0	59.7	63.8	66.4	66.6	65.4	62.5	57.4	54.7
		MCWB	51.1	50.3	51.1	52.3	55.8	59.4	62.0	62.3	61.5	59.4	54.8	52.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.4	54.8	55.6	56.8	61.0	64.5	66.2	67.2	67.0	64.3	60.4	56.6
		MCDB	56.1	56.5	59.5	61.7	66.6	69.1	70.1	72.4	71.7	68.7	62.2	57.8
	2%	WB	54.3	53.3	54.1	55.1	58.7	62.5	64.6	65.2	64.7	62.4	57.8	55.4
		MCDB	55.3	54.5	56.7	58.3	62.2	66.2	68.2	68.2	68.2	65.2	59.6	57.0
	5%	WB	53.0	52.0	52.8	53.9	57.5	61.2	63.5	64.0	63.4	61.1	56.4	54.2
		MCDB	54.2	53.5	55.2	56.8	60.5	64.3	66.9	66.7	66.3	63.4	58.3	55.9
	10%	WB	51.7	50.7	51.5	52.8	56.4	60.1	62.6	63.0	62.1	59.8	55.0	52.9
		MCDB	53.1	52.5	53.8	55.5	59.1	63.0	65.6	65.8	64.7	62.0	57.0	54.7
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	7.3	7.9	8.5	8.1	8.4	8.7	9.0	9.4	8.8	7.3	7.1
		MCDBR	7.3	8.3	10.5	11.5	11.1	11.1	10.9	11.2	12.5	10.8	8.9	7.1
		MCWBR	7.3	7.0	7.0	7.1	6.7	6.6	6.1	6.5	7.3	7.2	7.4	7.0
	5% WB	MCDBR	7.2	7.6	9.3	10.1	10.0	10.2	9.9	9.8	10.9	9.7	8.3	7.0
		MCWBR	7.7	7.3	6.9	7.0	6.5	6.6	6.3	6.4	7.0	7.1	7.7	7.4
(40) Clear-Sky Solar Irradiance	taub	0.339	0.353	0.380	0.387	0.391	0.383	0.385	0.373	0.364	0.360	0.340	0.332	(40)
	taud	2.385	2.371	2.314	2.317	2.353	2.402	2.403	2.455	2.461	2.453	2.419	2.352	(41)
	Ebn at Noon	226	251	263	273	276	279	276	275	266	246	224	210	(42)
	Edh at Noon	22	28	34	37	37	36	36	32	29	25	21	20	(43)
(44) All-Sky Solar Radiation	RadAvg	300	552	923	1458	1733	1878	1780	1525	1208	708	376	249	(44)
	RadStd	23	59	96	120	149	196	100	123	86	50	30	29	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page